

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 7

REVIEW Lecture 6:

- Direct Methods for solving linear algebraic equations

- LU decomposition/factorization

- Separates time-consuming elimination for $\mathbf{A}$ from that for $\mathbf{b} / \mathbf{B}$

$$\overline{\overline{\mathbf{A}}} = \overline{\overline{\mathbf{L}}} \cdot \overline{\overline{\mathbf{U}}} \longrightarrow \begin{aligned} \overline{\overline{\mathbf{L}}} \vec{y} &= \vec{b} \\ \overline{\overline{\mathbf{U}}} \vec{x} &= \vec{y} \end{aligned}$$

- Derivation, assuming no pivoting needed: $a_{ij} = \sum_{k=1}^{\min(i,j)} m_{ik} a_{kj}^{(k)}$
- Number of Ops: Same as for Gauss Elimination
- Pivoting: Use pivot element “pointer vector”
- Variations: Doolittle and Crout decompositions, Matrix Inverse

- Error Analysis for Linear Systems

- Matrix norms
- Condition Number for Perturbed RHS and LHS: $K(\overline{\overline{\mathbf{A}}}) = \left\| \overline{\overline{\mathbf{A}}}^{-1} \right\| \left\| \overline{\overline{\mathbf{A}}} \right\|$

- Special Matrices: Intro

ODAY (Lecture 7): Systems of Linear Equations III

Direct Methods

- Gauss Elimination
- LU decomposition/factorization
- Error Analysis for Linear Systems
- Special Matrices: LU Decompositions
 - Tri-diagonal systems: Thomas Algorithm
 - General Banded Matrices
 - Algorithm, Pivoting and Modes of storage
 - Sparse and Banded Matrices
 - Symmetric, positive-definite Matrices
 - Definitions and Properties, Choleski Decomposition

• Iterative Methods

- Jacobi's method
- Gauss-Seidel iteration
- Convergence

Reading Assignment

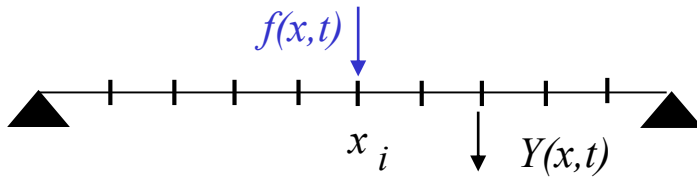
- Chapters 11 of “Chapra and Canale, Numerical Methods for Engineers, 2006/2010/2014.”
 - Any chapter on “Solving linear systems of equations” in references on CFD references provided. For example: chapter 5 of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”

Special Matrices

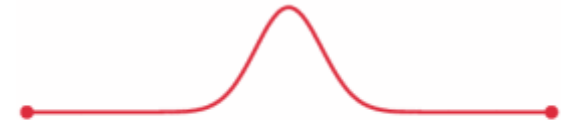
- **Certain Matrices have particular structures that can be exploited, i.e.**
 - Reduce number of ops and memory needs
- **Banded Matrices:**
 - Square banded matrix that has all elements equal to zero, excepted for a band around the main diagonal.
 - Frequent in engineering and differential equations:
 - Tri-diagonal Matrices
 - Wider bands for higher-order schemes
 - Gauss Elimination or LU decomposition inefficient because, if pivoting is not necessary, all elements outside of the band remain zero (but direct GE/LU would manipulate these zero elements anyway)
- **Symmetric Matrices**
- **Iterative Methods:**
 - Employ initial guesses, than iterate to refine solution
 - Can be subject to round-off errors

Special Matrices: Tri-diagonal Systems Example

Forced Vibration of a String



Example of a travelling pulse:



Consider the case of a Harmonic excitation

$$f(x,t) = -f(x) \cos(\omega t)$$

Applying Newton's law leads to the wave equation:
With separation of variables, one obtains the
equation for modal amplitudes, see eq. (1) below:

$$\begin{cases} Y_{tt} - c^2 Y_{xx} = f(x,t) \\ Y(x,t) = \tau(t) y(x) \end{cases}$$

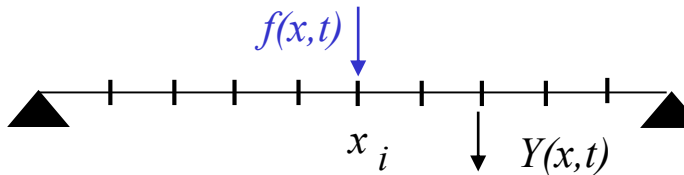
Differential Equation for the amplitude:

$$\frac{d^2 y}{dx^2} + k^2 y = f(x) \quad (1)$$

Boundary Conditions: $y(0) = 0$, $y(L) = 0$

Special Matrices: Tri-diagonal Systems

Forced Vibration of a String



Harmonic excitation

$$f(x,t) = f(x) \cos(\omega t)$$

Differential Equation:

$$\frac{d^2 y}{dx^2} + k^2 y = f(x) \quad (1)$$

Boundary Conditions:

$$y(0) = 0, \quad y(L) = 0$$

Finite Difference

$$\left. \frac{d^2 y}{dx^2} \right|_{x_i} \simeq \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} + O(h^2)$$

Discrete Difference Equations

$$y_{i-1} + ((kh)^2 - 2)y_i + y_{i+1} = f(x_i)h^2$$

Matrix Form:

$$\begin{bmatrix} (kh)^2 - 2 & 1 & \cdot & \cdot & \cdot & \cdot & 0 \\ 1 & (kh)^2 - 2 & 1 & & & & \\ \cdot & & \cdot & \cdot & & & \\ \cdot & & & \cdot & \cdot & & \\ \cdot & & 1 & (kh)^2 - 2 & 1 & & \\ \cdot & & & \cdot & \cdot & \cdot & \\ 0 & \cdot & \cdot & \cdot & \cdot & 1 & (kh)^2 - 2 \end{bmatrix} \bar{y} = \begin{Bmatrix} f(x_1)h^2 \\ \cdot \\ \cdot \\ f(x_i)h^2 \\ \cdot \\ \cdot \\ f(x_n)h^2 \end{Bmatrix}$$

Tridiagonal Matrix

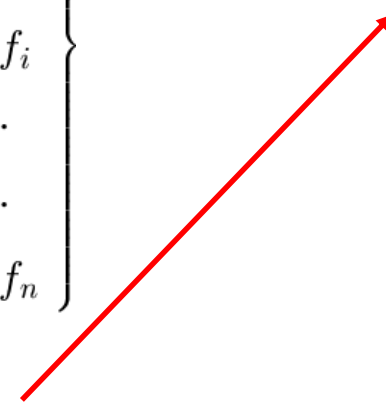
If $kh < 1$ or $kh > \sqrt{3}$ symmetric, negative or positive definite: No pivoting needed

Note: for $0 < kh < 1$ Negative definite $\Rightarrow$ Write: $\mathbf{A}' = -\mathbf{A}$ and $\bar{y}' = -\bar{y}$ to render matrix positive definite

Special Matrices: Tri-diagonal Systems

General Tri-diagonal Systems: Bandwidth of 3

$$\begin{bmatrix} a_1 & c_1 & \cdot & \cdot & \cdot & \cdot & 0 \\ b_2 & a_2 & c_2 & & & & \cdot \\ \cdot & & \cdot & \cdot & & & \cdot \\ \cdot & & b_i & a_i & c_i & & \cdot \\ \cdot & & & \cdot & \cdot & & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & b_n & a_n \end{bmatrix} \bar{\mathbf{x}} = \begin{Bmatrix} f_1 \\ \cdot \\ \cdot \\ f_i \\ \cdot \\ \cdot \\ f_n \end{Bmatrix}$$

$$\bar{\bar{\mathbf{L}}} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & 0 \\ \beta_2 & 1 & & & & & \cdot \\ \cdot & \cdot & \cdot & & & & \cdot \\ \cdot & \beta_i & 1 & & & & \cdot \\ \cdot & & \cdot & \cdot & & & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & \beta_n & 1 \end{bmatrix}$$


LU Decomposition

$$\bar{\bar{\mathbf{A}}} = \bar{\bar{\mathbf{L}}} \bar{\bar{\mathbf{U}}} \quad \begin{cases} \bar{\bar{\mathbf{L}}} \bar{\bar{\mathbf{y}}} = \bar{\bar{\mathbf{f}}} \\ \bar{\bar{\mathbf{U}}} \bar{\bar{\mathbf{x}}} = \bar{\bar{\mathbf{y}}} \end{cases}$$


$$\bar{\bar{\mathbf{U}}} = \begin{bmatrix} \alpha_1 & c_1 & \cdot & \cdot & \cdot & \cdot & 0 \\ & \alpha_2 & c_2 & & & & \cdot \\ \cdot & & \cdot & \cdot & & & \cdot \\ \cdot & & & \alpha_i & c_i & & \cdot \\ \cdot & & & \cdot & \cdot & & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & & \alpha_n \end{bmatrix}$$

Three steps for LU scheme:

1. Decomposition (GE): $a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)}, \quad m_{ik} = a_{ik}^{(k)} / a_{kk}^{(k)}$
2. Forward substitution $\bar{\bar{\mathbf{L}}} \bar{\bar{\mathbf{y}}} = \bar{\bar{\mathbf{f}}}$
3. Backward substitution $\bar{\bar{\mathbf{U}}} \bar{\bar{\mathbf{x}}} = \bar{\bar{\mathbf{y}}}$

Special Matrices: Tri-diagonal Systems

Thomas Algorithm

By identification with the general LU decomposition, $a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik}a_{kj}^{(k)}$, $m_{ik} = a_{ik}^{(k)} / a_{kk}^{(k)}$

one obtains,

$$\bar{\bar{\mathbf{L}}} = \begin{bmatrix} 1 & . & . & . & . & 0 \\ \beta_2 & 1 & & & & . \\ . & . & . & & & . \\ . & \beta_i & 1 & & & . \\ . & . & . & . & & . \\ 0 & . & . & . & . & \beta_n & 1 \end{bmatrix}$$

1. Factorization/Decomposition

$$\alpha_1 = a_1$$

$$\beta_k = \frac{b_k}{\alpha_{k-1}}, \quad \alpha_k = a_k - \beta_k c_{k-1}, \quad k = 2, 3, \dots, n$$

2. Forward Substitution

$$y_1 = f_1, \quad y_i = f_i - \beta_i y_{i-1}, \quad i = 2, 3, \dots, n$$

3. Back Substitution

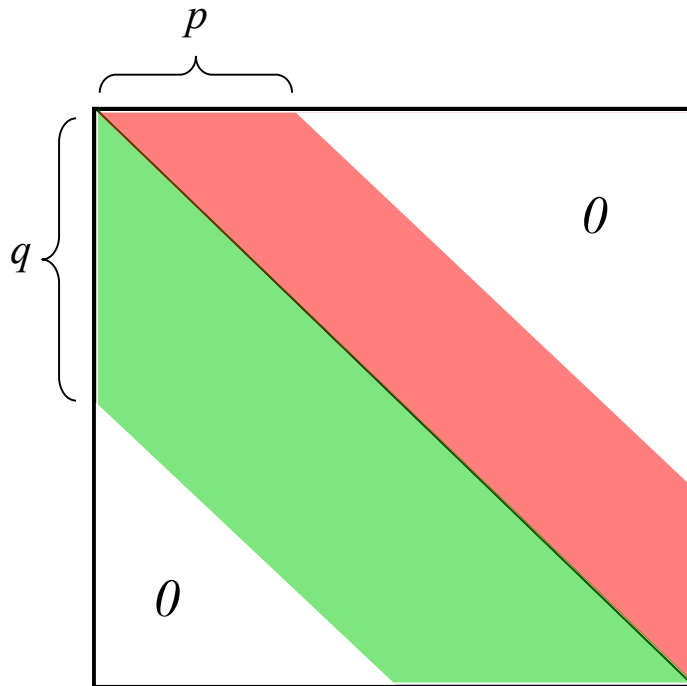
$$x_n = \frac{y_n}{\alpha_n}, \quad x_i = \frac{y_i - c_i x_{i+1}}{\alpha_i}, \quad i = n-1, \dots, 1$$

$$\bar{\bar{\mathbf{U}}} = \begin{bmatrix} \alpha_1 & c_1 & . & . & . & . & 0 \\ . & \alpha_2 & c_2 & & & & . \\ . & . & . & . & & & . \\ . & . & \alpha_i & c_i & & & . \\ . & . & . & . & . & & . \\ 0 & . & . & . & . & . & \alpha_n \end{bmatrix}$$

Number of Operations: Thomas Algorithm

LU Factorization:	3*(n-1) operations
Forward substitution:	2*(n-1) operations
Back substitution:	3*(n-1)+1 operations
Total:	8*(n-1) ~ O(n) operations

Special Matrices: General, Banded Matrix



p super-diagonals
 q sub-diagonals
 $w = p + q + 1$ bandwidth

General Banded Matrix ($p \neq q$)

$$\left. \begin{array}{l} j > i + p \\ i > j + q \end{array} \right\} a_{ij} = 0$$

Banded Symmetric Matrix ($p = q = b$)

$$\begin{aligned} a_{ij} &= a_{ji}, \quad |i - j| \leq b \\ a_{ij} &= a_{ji} = 0, \quad |i - j| > b \end{aligned}$$

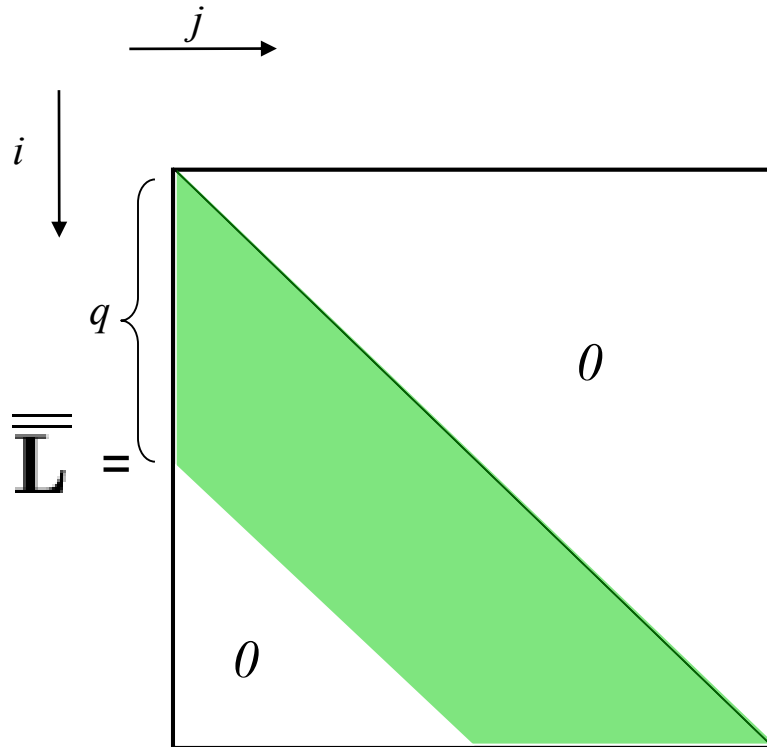
$w = 2b + 1$ is called the bandwidth
 b is the half-bandwidth

Special Matrices:

General, Banded Matrix

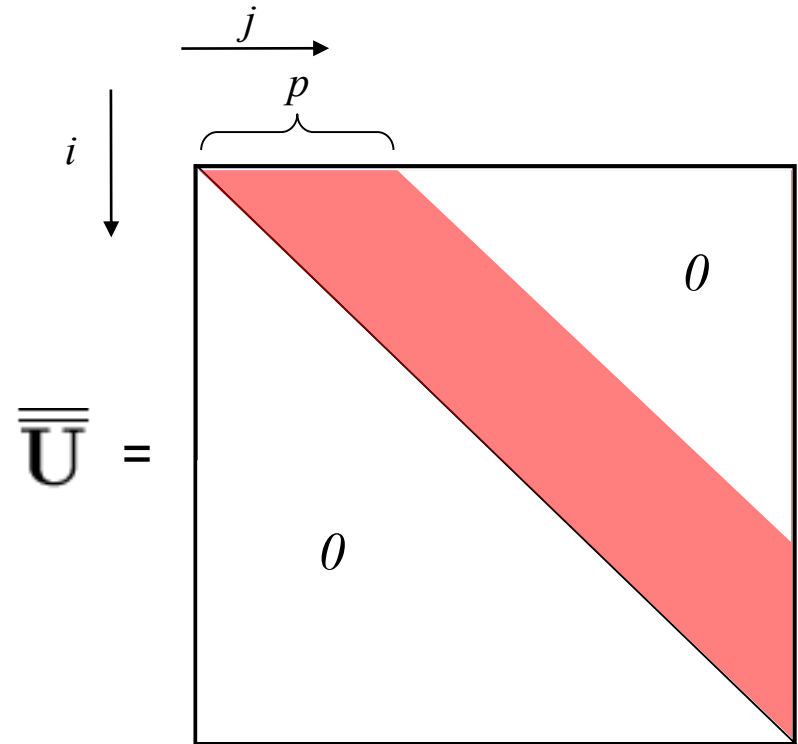
LU Decomposition via Gaussian Elimination

If **No Pivoting**: the zeros are preserved



$$m_{ij} = \frac{a_{ij}^{(j)}}{a_{jj}^{(j)}} = 0 \quad \text{if } j > i \text{ or } \underline{i > j + q}$$

(as gen. case) (banded)



$$u_{ij} = a_{ij}^{(i)} = a_{ij}^{(i-1)} - m_{i,i-1} a_{i-1,j}^{(i-1)}$$

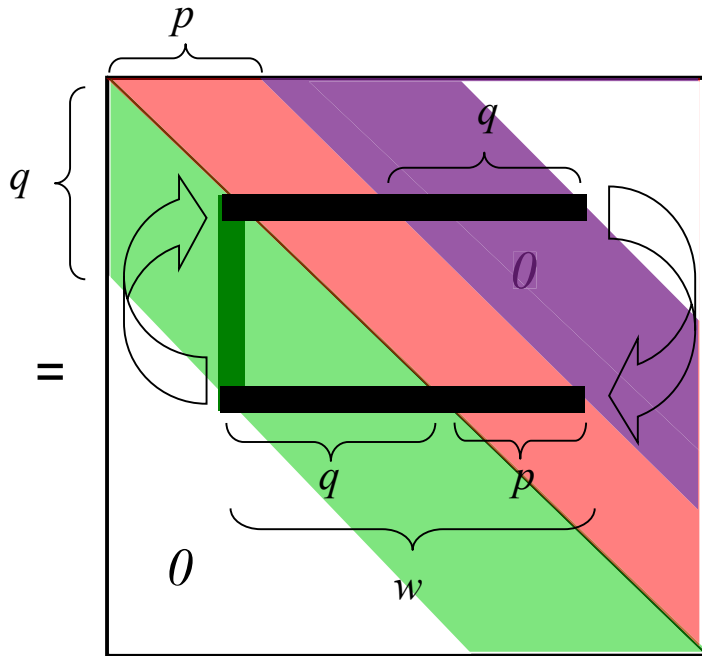
$$u_{ij} = 0 \quad \text{if } i > j \text{ or } \underline{j > i + p}$$

(as gen. case) (banded)

Special Matrices: General, Banded Matrix

LU Decomposition via Gaussian Elimination
With **Partial Pivoting** (by rows):

Consider pivoting the 2 rows as below:



Then, the bandwidth of L remains unchanged,

$$m_{ij} = 0 \quad \text{if} \quad j > i \quad \text{or} \quad \underline{i > j + q}$$

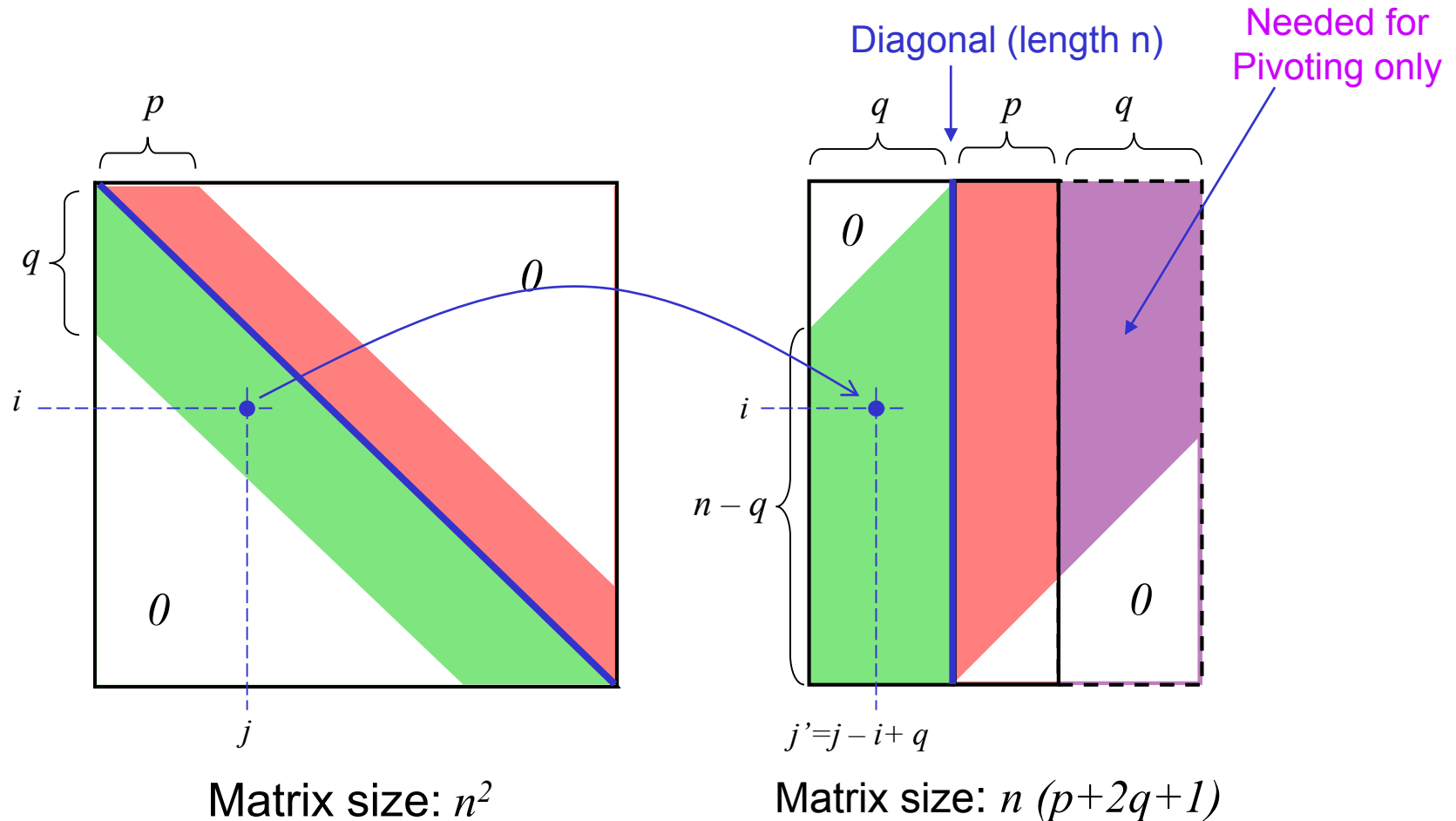
but the bandwidth of U becomes as that of A

$$u_{ij} = 0 \quad \text{if} \quad i > j \quad \text{or} \quad \underline{j > i + p + q}$$

$$w = p + 2q + 1 \text{ bandwidth}$$

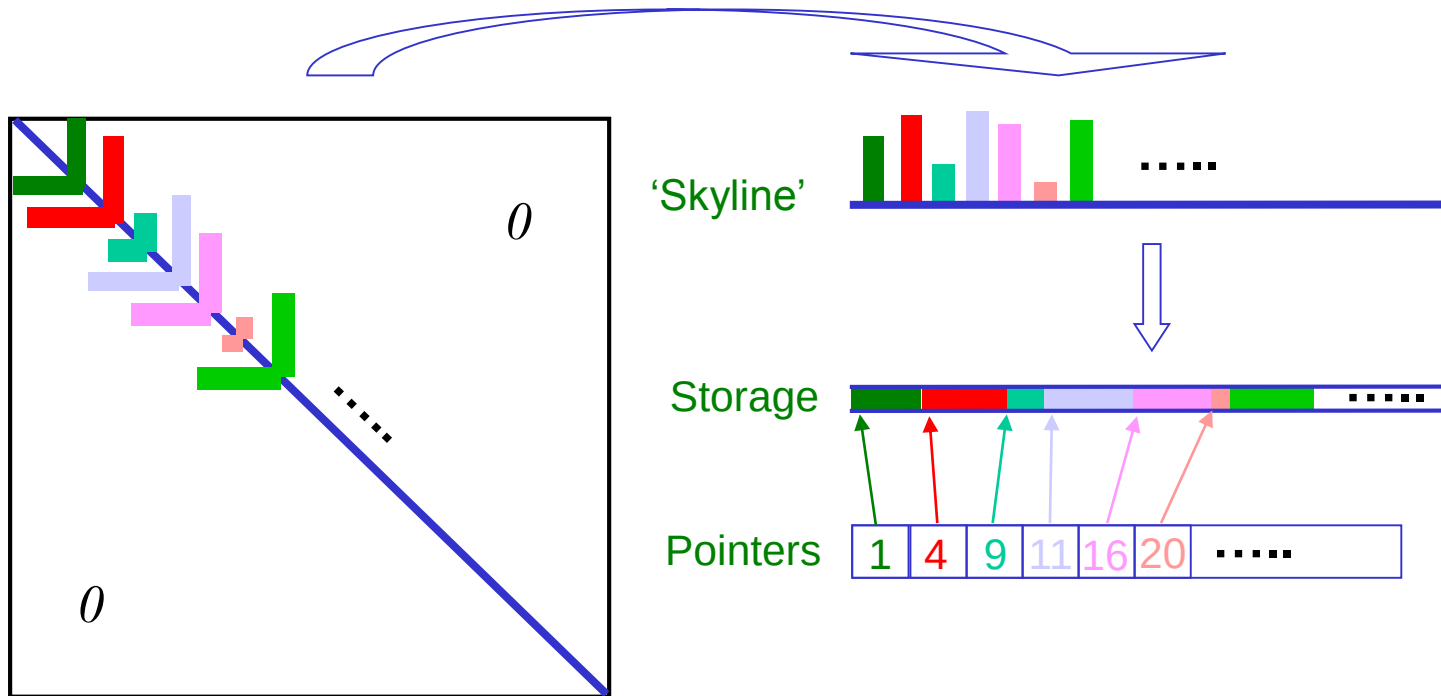
Special Matrices: General, Banded Matrix

Compact Storage



Special Matrices: Sparse and Banded Matrix

‘Skyline’ Systems (typically for symmetric matrices)



Skyline storage applicable when no pivoting is needed, e.g. for banded, symmetric, and positive definite matrices: FEM and FD methods. Skyline solvers are usually based on Cholesky factorization (which preserves the skyline)

Special Matrices:

Symmetric (Positive-Definite) Matrix

Symmetric Coefficient Matrices:

- If no pivoting, the matrix remains symmetric after Gauss Elimination/LU decompositions

Proof: Show that if $a_{ij}^{(k)} = a_{ji}^{(k)}$ then $a_{ij}^{(k+1)} = a_{ji}^{(k+1)}$ using:

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik}a_{kj}^{(k)}, \quad m_{ik} = a_{ik}^{(k)} / a_{kk}^{(k)}$$

- Gauss Elimination symmetric (use only the upper triangular portion of **A**):

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)}$$

$$m_{ik} = \frac{a_{ki}^{(k)}}{a_{kk}^{(k)}}, \quad i = k+1, k+2, \dots, n \quad j = i, i+1, \dots, n$$

- About half the total number of ops than full GE

Special Matrices:

Symmetric, Positive Definite Matrix

1. Sylvester Criterion:

A symmetric matrix is Positive Definite if and only if:

$\det(\mathbf{A}_k) > 0$ for $k=1,2,\dots,n$, where $\mathbf{A}_k$ is matrix of k first lines/columns

Symmetric Positive Definite matrices frequent in engineering

2. For a symmetric positive definite $\mathbf{A}$, one thus has the following properties

a) The maximum elements of $\mathbf{A}$ are on the main diagonal

b) For a Symmetric, Positive Definite $\mathbf{A}$: No pivoting needed

c) The elimination is stable: $|a_{ii}^{(k+1)}| \leq 2 |a_{ii}^{(k)}|$. To show this, use $a_{kj}^2 \leq a_{kk} a_{jj}$ in

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)}$$

$$m_{ik} = \frac{a_{ki}^{(k)}}{a_{kk}^{(k)}}, \quad i = k+1, k+2, \dots, n \quad j = i, i+1, \dots, n$$

Special Matrices:

Symmetric, Positive Definite Matrix

The general GE $\left\{ \begin{array}{l} a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)} \\ m_{ik} = \frac{a_{ki}^{(k)}}{a_{kk}^{(k)}}, \quad i = k+1, k+2, \dots, n \quad j = i, i+1, \dots, n \end{array} \right.$

$$a_{ij} = \sum_{k=1}^{\min(i,j)} m_{ik} a_{kj}^{(k)}$$

becomes:

$$\overline{\overline{\mathbf{A}}} = \overline{\overline{\mathbf{L}}} \overline{\overline{\mathbf{U}}} = \overline{\overline{\mathbf{U}}}^\dagger \overline{\overline{\mathbf{U}}}$$

Choleski Factorization

$$\overline{\overline{\mathbf{U}}}^\dagger = [m_{ij}]$$

Complex Conjugate

where

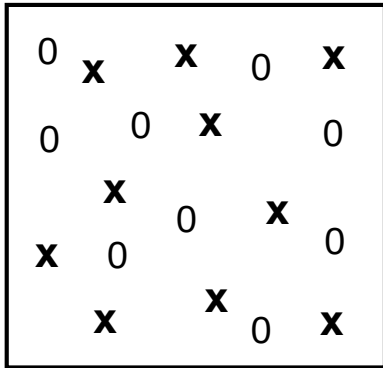
$$\left. \begin{array}{l} m_{kk} = \left(a_{kk} - \sum_{\ell=1}^{k-1} m_{k\ell} \overline{m_{k\ell}} \right)^{1/2} \\ m_{ik} = \frac{a_{ik} - \sum_{\ell=1}^{k-1} m_{i\ell} \overline{m_{k\ell}}}{m_{kk}}, \quad i = k+1, \dots, n \end{array} \right\} k = 1, \dots, n$$

No pivoting needed

† Complex Conjugate and Transpose

Linear Systems of Equations: Iterative Methods

Sparse (large) Full-bandwidth Systems (frequent in practice)



Iterative Methods are then efficient

Analogous to iterative methods obtained for roots of equations, i.e. Open Methods: Fixed-point, Newton-Raphson, Secant

Example of Iteration equation

$$\mathbf{A} \mathbf{x} = \mathbf{b} \Rightarrow \mathbf{A} \mathbf{x} - \mathbf{b} = 0$$

$$\mathbf{x} = \mathbf{x} + \mathbf{A} \mathbf{x} - \mathbf{b} \Rightarrow$$

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \mathbf{A} \mathbf{x}^k - \mathbf{b} = (\mathbf{A} + \mathbf{I}) \mathbf{x}^k - \mathbf{b}$$

General Stationary Iteration Formula

$$\mathbf{x}^{k+1} = \mathbf{B} \mathbf{x}^k + \mathbf{c} \quad k = 0, 1, 2, \dots$$

Compatibility condition for $\mathbf{A}\mathbf{x}=\mathbf{b}$ to be the solution:

$$\left. \begin{array}{l} \text{Write } \mathbf{c} = \mathbf{C} \mathbf{b} \\ \mathbf{A}^{-1} \mathbf{b} = \mathbf{B} \mathbf{A}^{-1} \mathbf{b} + \mathbf{C} \mathbf{b} \end{array} \right\} \Rightarrow (\mathbf{I} - \mathbf{B}) \mathbf{A}^{-1} = \mathbf{C} \quad \text{or} \quad \mathbf{B} = \mathbf{I} - \mathbf{C} \mathbf{A}$$

ps: $\mathbf{B}$ and $\mathbf{c}$ could be function of k (non-stationary)

Linear Systems of Equations: Iterative Methods

Convergence

Convergence

$$\|\bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}}\| \rightarrow 0 \text{ for } k \rightarrow \infty$$

Iteration – Matrix form

$$\bar{\mathbf{x}}^{(k+1)} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}}^{(k)} + \bar{\mathbf{c}}, \quad k = 0, \dots$$

Convergence Analysis

$$\bar{\mathbf{x}}^{(k+1)} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}}^{(k)} + \bar{\mathbf{c}}$$

$$\bar{\mathbf{x}} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}} + \bar{\mathbf{c}}$$

$$\begin{aligned} \Rightarrow \quad \bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}} &= \bar{\bar{\mathbf{B}}} (\bar{\mathbf{x}}^{(k)} - \bar{\mathbf{x}}) \\ &= \bar{\bar{\mathbf{B}}} \cdot \bar{\bar{\mathbf{B}}} (\bar{\mathbf{x}}^{(k-1)} - \bar{\mathbf{x}}) \\ &\quad \cdot \\ &= \bar{\bar{\mathbf{B}}}^{k+1} (\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}) \end{aligned}$$

$$\|\bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}}\| \leq \|\bar{\bar{\mathbf{B}}}^{k+1}\| \|\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}\| \leq \|\bar{\bar{\mathbf{B}}}\|^{k+1} \|\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}\|$$

Sufficient Condition for Convergence:

$$\|\bar{\bar{\mathbf{B}}}\| < 1$$

$\|B\| < 1$ for a chosen matrix norm
Infinite norm often used in practice

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$$

“Maximum Column Sum”

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$$

“Maximum Row Sum”

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{1/2}$$

“The Frobenius norm” (also called Euclidean norm)”, which for matrices differs from:

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^*A)}$$

“The l-2 norm” (also called spectral norm)

Linear Systems of Equations: Iterative Methods

Convergence: Necessary and Sufficient Condition

Convergence

$$\|\bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}}\| \rightarrow 0 \text{ for } k \rightarrow \infty$$

Iteration – Matrix form

$$\bar{\mathbf{x}}^{(k+1)} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}}^{(k)} + \bar{\mathbf{c}}, \quad k = 0, \dots$$

Convergence Analysis

$$\bar{\mathbf{x}}^{(k+1)} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}}^{(k)} + \bar{\mathbf{c}}$$

$$\bar{\mathbf{x}} = \bar{\bar{\mathbf{B}}} \bar{\mathbf{x}} + \bar{\mathbf{c}}$$

$$\begin{aligned} \Rightarrow \quad \bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}} &= \bar{\bar{\mathbf{B}}} (\bar{\mathbf{x}}^{(k)} - \bar{\mathbf{x}}) \\ &= \bar{\bar{\mathbf{B}}} \cdot \bar{\bar{\mathbf{B}}} (\bar{\mathbf{x}}^{(k-1)} - \bar{\mathbf{x}}) \\ &\quad \cdot \\ &= \bar{\bar{\mathbf{B}}}^{k+1} (\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}) \end{aligned}$$

$$\|\bar{\mathbf{x}}^{(k+1)} - \bar{\mathbf{x}}\| \leq \|\bar{\bar{\mathbf{B}}}^{k+1}\| \|\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}\| \leq \|\bar{\bar{\mathbf{B}}}\|^{k+1} \|\bar{\mathbf{x}}^{(0)} - \bar{\mathbf{x}}\|$$

Necessary and Sufficient Condition for Convergence:

Spectral radius of $\mathbf{B}$ is smaller than one: $\rho(\mathbf{B}) = \max_{i=1 \dots n} |\lambda_i| < 1$, where $\lambda_i = \text{eigenvalue}(\mathbf{B}_{n \times n})$

(proof: use eigendecomposition of $\mathbf{B}$)

(This ensures $\|\mathbf{B}\| < 1$)

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